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(57) Abstract :
The present invention discloses Swasthya 24/7— a solar-powered, IoT-enabled, autonomous medicine dispensing kiosk integrated with a Hospital-Connected Cloud Prescription Database, a Cloud-Integrated RFID Smart Health Card system, and an AI-assisted Proximity-Triggered Ambient Voice Guidance system. The invention is conceived to permanently resolve the critical last-mile healthcare access deficit experienced by rural, tribal, semi-urban, and underserved communities across India and other developing nations, where pharmacies, qualified pharmacists, and healthcare facilities are either absent, geographically inaccessible, or unavailable outside business hours. The Swasthya 24/7 operates continuously — 24 hours a day, 365 days a year — without requiring a full-time pharmacist or continuous internet connectivity. When a patient visits any partnered hospital or PHC, the attending doctor uploads their prescription directly into the Swasthya 24/7 Cloud Health Database. Each registered patient is then issued an RFID Smart Health Card that, upon authentication at any kiosk nationwide, instantly retrieves their live encrypted medical profile — including all hospital-uploaded prescriptions, chronic illness records, dosage histories, allergy data, and refill eligibility — without the patient carrying any paper document. A core accessibility innovation is the Proximity-Triggered Ambient AI Voice Guidance System: when a person enters within 2–5 metres of the kiosk, the AI automatically begins narrating step-by-step instructions in the local regional language — guiding even a completely illiterate, first-time user through the entire medicine collection process without any touch input required to begin, analogous to metro rail public announcement systems that continuously guide passengers before they reach their destination. The kiosk then permits dispensing of only prescription-approved medicines, generates real-time drug interaction alerts via voice and visual interfaces, and maintains a tamper-evident dispensing audit log. The system is powered by a 300W solar panel array with 150Ah battery backup, ensuring uninterrupted operation during power outages. A mechanical carousel-based automated dispensing mechanism, controlled by servo motors interfaced with a Raspberry Pi 4 microcontroller, executes medicine release with precision. Payments are supported via cash, UPI, debit/credit card, and Aadhaar Pay — ensuring access for all users regardless of digital literacy or banking status. An emergency override module enables authorized PHC doctors to remotely grant access to critical medicines. The invention is scalable across Primary Health Centres, Panchayat offices, rural schools, border posts, tribal welfare centres, and remote geographies globally.

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